



Solution of Triangle

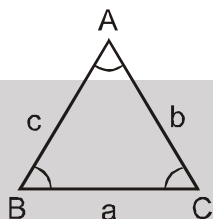
According to most accounts, geometry was first discovered among the Egyptians, taking its origin from the measurement of areas. For they found it necessary by reason of the flooding of the Nile, which wiped out everybody's proper boundaries. Nor is there anything surprising in that the discovery both of this and of the other sciences should have had its origin in a practical need, since everything which is in process of becoming progresses from the imperfect to the perfect.

.....Proclus

Sine Rule :

In any triangle ABC, the sines of the angles are proportional to the opposite sides

$$\text{i.e. } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$



Example # 1 : How many triangles can be constructed with the data : $a = 5$, $b = 7$, $\sin A = 3/4$

Solution : Since $\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow \frac{5}{3/4} = \frac{7}{\sin B}$
 $\Rightarrow \sin B = \frac{21}{20} > 1$ not possible
 $\therefore$ no triangle can be constructed.

Example # 2 : If in a triangle ABC, $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, then show that a^2, b^2, c^2 are in A.P.

Solution : We have $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$
 $\Rightarrow \sin(B+C) \sin(B-C) = \sin(A+B) \sin(A-B) \Rightarrow \sin^2 B - \sin^2 C = \sin^2 A - \sin^2 B$
 $\Rightarrow b^2 - c^2 = a^2 - b^2 \Rightarrow a^2, b^2, c^2$ are in A.P.

Self Practice Problems :

- (1) In a $\triangle ABC$, the sides a, b and c are in A.P., then prove that $\left(\tan \frac{A}{2} + \tan \frac{C}{2} \right) : \cot \frac{B}{2} = 2 : 3$
- (2) If the angles of $\triangle ABC$ are in the ratio $1 : 2 : 3$, then find the ratio of their corresponding sides
- (3) In a $\triangle ABC$ prove that $\frac{c}{a-b} = \frac{\tan \frac{A}{2} + \tan \frac{B}{2}}{\tan \frac{A}{2} - \tan \frac{B}{2}}$.

Ans. (2) $1 : \sqrt{3} : 2$

Cosine Formula :

In any $\triangle ABC$

$$(i) \quad \cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad \text{or} \quad a^2 = b^2 + c^2 - 2bc \cos A = b^2 + c^2 + 2bc \cos(B+C)$$

$$(ii) \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$(iii) \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$





Example # 3 : In a triangle ABC, A, B, C are in A.P. Show that $2\cos\left(\frac{A-C}{2}\right) = \frac{a+c}{\sqrt{a^2-ac+c^2}}$.

Solution : $A + C = 2B \Rightarrow A + B + C = 3B \Rightarrow B = 60^\circ$

$$\therefore \cos 60^\circ = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow a^2 - ac + c^2 = b^2$$

$$\begin{aligned} \Rightarrow \frac{a+c}{\sqrt{a^2-ac+c^2}} &= \frac{a+c}{b} = \left[\frac{\sin A + \sin C}{\sin B} \right] = \frac{2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)}{\sin B} \\ &= 2\cos\frac{A-C}{2} \quad (\because A+C=2B) \end{aligned}$$

Example # 4 : In a $\triangle ABC$, prove that $a(b\cos C - c\cos B) = b^2 - c^2$

Solution : Since $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ & $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$$\begin{aligned} \therefore \text{L.H.S.} &= a \left\{ b \left(\frac{a^2 + b^2 - c^2}{2ab} \right) - c \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \right\} \\ &= \frac{a^2 + b^2 - c^2}{2} - \frac{(a^2 + c^2 - b^2)}{2} = (b^2 - c^2) = \text{R.H.S.} \end{aligned}$$

Hence L.H.S. = R.H.S.

Proved

Example # 5 : The sides of $\triangle ABC$ are $AB = \sqrt{13}$ cm, $BC = 4\sqrt{3}$ cm and $CA = 7$ cm. Then find the value of $\sin\theta$ where θ is the smallest angle of the triangle.

Solution : Angle opposite to AB is smallest. Therefore,

$$\cos\theta = \frac{49 + 48 - 13}{2 \cdot 7 \cdot 4\sqrt{3}} = \frac{\sqrt{3}}{2} \Rightarrow \sin\theta = \frac{1}{2}$$

Self Practice Problems :

(4) If in a triangle ABC, $3\sin A = 6\sin B = 2\sqrt{3}\sin C$, Then find the angle A.

(5) If two sides a, b and angle A be such that two triangles are formed, then find the sum of two values of the third side.

Ans. (4) 90° (5) $2b\cos A$

Projection Formula :

In any $\triangle ABC$

(i) $a = b\cos C + c\cos B$

(ii) $b = c\cos A + a\cos C$

(iii) $c = a\cos B + b\cos A$

Example # 6 : If in a $\triangle ABC$, $c\cos^2\frac{A}{2} + a\cos^2\frac{C}{2} = \frac{3b}{2}$, then show that a, b, c are in A.P.

Solution : $c(1 + \cos A) + a(1 + \cos C) = 3b$
 $\Rightarrow a + c + (c\cos A + a\cos C) = 3b$
 $\Rightarrow a + c + b = 3b$
 $\Rightarrow a + c = 2b$

Example # 7 : In a $\triangle ABC$, prove that $(b+c)\cos A + (c+a)\cos B + (a+b)\cos C = a+b+c$.

Solution : $\therefore \text{L.H.S.} = (b+c)\cos A + (c+a)\cos B + (a+b)\cos C$

$$\begin{aligned} &= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C \\ &= (b\cos A + a\cos B) + (c\cos A + a\cos C) + (c\cos B + b\cos C) \\ &= a + b + c \\ &= \text{R.H.S.} \end{aligned}$$

Hence L.H.S. = R.H.S.

Proved





Self Practice Problems :

- (6) The roots of $x^2 - 2\sqrt{3}x + 2 = 0$ represent two sides of a triangle. If the angle between them is $\frac{\pi}{3}$, then find the perimeter of triangle.
- (7) In a triangle ABC, if $\cos A + \cos B + \cos C = 3/2$, then show that the triangle is an equilateral triangle.
- (8) In a $\triangle ABC$, prove that $\frac{\cos A}{c \cos B + b \cos C} + \frac{\cos B}{a \cos C + c \cos A} + \frac{\cos C}{a \cos B + b \cos A} = \frac{a^2 + b^2 + c^2}{2abc}$.

Ans. (6) $2\sqrt{3} + \sqrt{6}$

Napier's Analogy - tangent rule :

In any $\triangle ABC$

$$\begin{aligned} \text{(i)} \quad \tan \frac{B-C}{2} &= \frac{b-c}{b+c} \cot \frac{A}{2} & \text{(ii)} \quad \tan \frac{C-A}{2} &= \frac{c-a}{c+a} \cot \frac{B}{2} \\ \text{(iii)} \quad \tan \frac{A-B}{2} &= \frac{a-b}{a+b} \cot \frac{C}{2} \end{aligned}$$

Example # 8 : Find the unknown elements of the $\triangle ABC$ in which $a = \sqrt{3} + 1$, $b = \sqrt{3} - 1$, $C = 90^\circ$.

Solution :

$\therefore a = \sqrt{3} + 1, b = \sqrt{3} - 1, C = 90^\circ$

$\therefore A + B + C = 180^\circ$

$\therefore A + B = 90^\circ$ (i)

$\therefore$ From law of tangent, we know that $\tan \left(\frac{A-B}{2} \right) = \frac{a-b}{a+b} \cot \frac{C}{2}$

$= \frac{(\sqrt{3}+1) - (\sqrt{3}-1)}{(\sqrt{3}+1) + (\sqrt{3}-1)} \cot 45^\circ = \frac{2}{2\sqrt{3}} \cot 45^\circ \Rightarrow \tan \left(\frac{A-B}{2} \right) = \frac{1}{\sqrt{3}}$

$\therefore \frac{A-B}{2} = \frac{\pi}{6}$

$\Rightarrow A - B = \frac{\pi}{3}$ (ii)

From equation (i) and (ii), we get $A = \frac{5\pi}{12}$ and $B = \frac{\pi}{12}$

Now, $c = \sqrt{a^2 + b^2} = 2\sqrt{2}$

$\therefore c = 2\sqrt{2}, A = \frac{5\pi}{12}, B = \frac{\pi}{12}$ **Ans.**

Self Practice Problems :

- (9) In a $\triangle ABC$ if $b = 3, c = 5$ and $\cos (B - C) = \frac{7}{25}$, then find the value of $\sin \frac{A}{2}$.
- (10) If in a $\triangle ABC$, we define $x = \tan \left(\frac{B-C}{2} \right) \tan \frac{A}{2}$, $y = \tan \left(\frac{C-A}{2} \right) \tan \frac{B}{2}$ and $z = \tan \left(\frac{A-B}{2} \right) \tan \frac{C}{2}$, then show that $x + y + z = -xyz$.

Ans. (9) $\frac{1}{\sqrt{10}}$





Trigonometric Functions of Half Angles :

- (i) $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$, $\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$
- (ii) $\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$, $\cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$
- (iii) $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{\Delta}{s(s-a)} = \frac{(s-b)(s-c)}{\Delta}$, where $s = \frac{a+b+c}{2}$ is semi perimeter and Δ is the area of triangle.
- (iv) $\sin A = \frac{2}{bc} \sqrt{s(s-a)(s-b)(s-c)} = \frac{2\Delta}{bc}$

Area of Triangle (Δ)

$$\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \sqrt{s(s-a)(s-b)(s-c)}$$

Example # 9 : If p_1, p_2, p_3 are the altitudes of a triangle ABC from the vertices A, B, C and Δ is the area of the triangle, then show that $p_1^{-1} + p_2^{-1} + p_3^{-1} = \frac{s-c}{\Delta}$

Solution : We have

$$\begin{aligned} \frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} &= \frac{a}{2\Delta} + \frac{b}{2\Delta} + \frac{c}{2\Delta} \\ &= \frac{a+b+c}{2\Delta} = \frac{2(s-c)}{2\Delta} = \frac{s-c}{\Delta} \end{aligned}$$

Example # 10 : In a ΔABC if $b \sin C (b \cos C + c \cos B) = 64$, then find the area of the ΔABC .

Solution : $\therefore b \sin C (b \cos C + c \cos B) = 64$ (i) given

$\therefore$ From **projection rule**, we know that
 $a = b \cos C + c \cos B$ put in (i), we get
 $ab \sin C = 64$ (ii)

$\therefore \Delta = \frac{1}{2} ab \sin C$ $\therefore$ from equation (ii), we get
 $\therefore \Delta = 32$ sq. unit

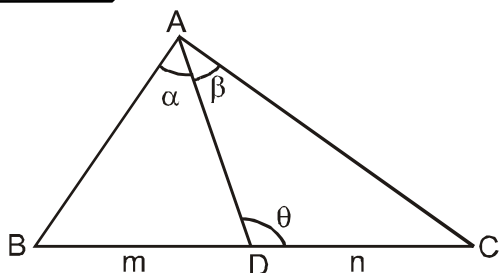
Example # 11 : If A, B, C are the angle of a triangle, then prove that $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \frac{s^2}{\Delta}$

Solution :

$$\begin{aligned} &\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} \\ &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-b)}{(s-c)(s-a)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ &= \frac{\sqrt{s(s-a+s-b+s-c)}}{\sqrt{(s-a)(s-b)(s-c)}} = \frac{s}{\Delta} (3s-2s) = \frac{s^2}{\Delta} \end{aligned}$$

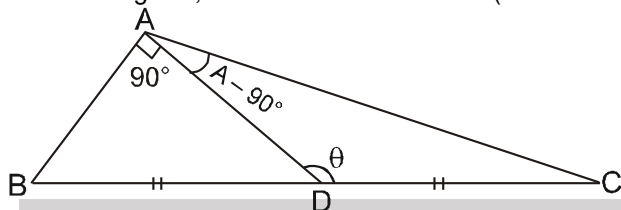
m - n Rule : In any triangle ABC if D be any point on the base BC, such that $BD : DC :: m : n$ and if $\angle BAD = \alpha$, $\angle DAC = \beta$, $\angle CDA = \theta$, then
 $(m+n) \cot \theta = m \cot \alpha + n \cot \beta$
 $n \cot B + m \cot C$





Example # 12 : In a $\triangle ABC$. AD divides BC in the ratio 2 : 1 such that at $\angle BAD = 90^\circ$ then prove that $\tan A + 3 \tan B = 0$

Solution : From the figure, we see that $\theta = 90^\circ + B$ (as θ is external angle of $\triangle ABD$)



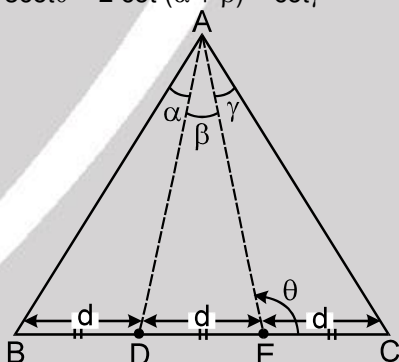
Now if we apply **m-n rule** in $\triangle ABC$, we get
 $(2 + 1) \cot (90^\circ + B) = 2 \cdot \cot 90^\circ - 1 \cdot \cot (A - 90^\circ)$
 $\Rightarrow -3 \tan B = \cot (90^\circ - A)$
 $\Rightarrow -3 \tan B = \tan A$
 $\Rightarrow \tan A + 3 \tan B = 0$ **Hence proved.**

Example # 13 : The base of a $\triangle$ is divided into three equal parts. If α, β, γ be the angles subtended by these parts at the vertex, prove that :

$$(\cot \alpha + \cot \beta)(\cot \beta + \cot \gamma) = 4 \operatorname{cosec}^2 \theta$$

Solution : Let point D and E divides the base BC into three equal parts i.e. $BD = DE = EC = d$ (Let) and let α, β and γ be the angles subtended by BD, DE and EC respectively at their opposite vertex. Now in $\triangle ABC$

$$\begin{aligned} \therefore BE : EC &= 2d : d = 2 : 1 \\ \therefore \text{from m-n rule, we get} \\ (2 + 1) \cot \theta &= 2 \cot (\alpha + \beta) - \cot \gamma \\ \Rightarrow 3 \cot \theta &= 2 \cot (\alpha + \beta) - \cot \gamma \quad \dots\dots(i) \end{aligned}$$



again

$$\begin{aligned} \therefore \text{in } \triangle ADC \\ \therefore DE : EC &= d : d = 1 : 1 \\ \therefore \text{if we apply m-n rule in } \triangle ADC, \text{ we get} \\ (1 + 1) \cot \theta &= 1 \cdot \cot \beta - 1 \cdot \cot \gamma \\ 2 \cot \theta &= \cot \beta - \cot \gamma \quad \dots\dots(ii) \end{aligned}$$

$$\text{from (i) and (ii), we get } \frac{3 \cot \theta}{2 \cot \theta} = \frac{2 \cot (\alpha + \beta) - \cot \gamma}{\cot \beta - \cot \gamma}$$

$$\Rightarrow 3 \cot \beta - 3 \cot \gamma = 4 \cot (\alpha + \beta) - 2 \cot \gamma$$





$$\begin{aligned}
 &\Rightarrow 3\cot\beta - \cot\gamma = 4 \cot(\alpha + \beta) \\
 &\Rightarrow 3\cot\beta - \cot\gamma = 4 \left\{ \frac{\cot\alpha \cot\beta - 1}{\cot\beta + \cot\alpha} \right\} \\
 &\Rightarrow 3\cot^2\beta + 3\cot\alpha \cot\beta - \cot\beta \cot\gamma - \cot\alpha \cot\gamma = 4 \cot\alpha \cot\beta - 4 \\
 &\Rightarrow 4 + 3\cot^2\beta = \cot\alpha \cot\beta + \cot\beta \cot\gamma + \cot\alpha \cot\gamma \\
 &\Rightarrow 4 + 4\cot^2\beta = \cot\alpha \cot\beta + \cot\alpha \cot\gamma + \cot\beta \cot\gamma + \cot^2\beta \\
 &\Rightarrow 4(1 + \cot^2\beta) = (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma) \\
 &\Rightarrow (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma) = 4\operatorname{cosec}^2\beta
 \end{aligned}$$

Self Practice Problems :

- (11) In a $\triangle ABC$, the median to the side BC is of length $\frac{1}{\sqrt{11-6\sqrt{3}}}$ unit and it divides angle A into the angles of 30° and 45° . Prove that the side BC is of length 2 unit.

Radius of Circumcircle :

If R be the circumradius of $\triangle ABC$, then $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}$

Example # 14 : In a $\triangle ABC$, prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

Solution : In a $\triangle ABC$, we know that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$
 and $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$
 $= \frac{4abc}{8R^3} = \frac{16\Delta R}{8R^3} = \frac{2\Delta}{R^2}$

Example # 15 : In a $\triangle ABC$ if $a = 22$ cm, $b = 28$ cm and $c = 36$ cm, then find its circumradius.

Solution : $\therefore R = \frac{abc}{4\Delta} \dots\dots(i)$
 $\therefore \Delta = \sqrt{s(s-a)(s-b)(s-c)}$
 $\therefore s = \frac{a+b+c}{2} = 43$ cm
 $\therefore \Delta = \sqrt{43 \times 21 \times 15 \times 7} = 21\sqrt{215}$
 $\therefore R = \frac{22 \times 28 \times 36}{4 \times 21\sqrt{215}} = \frac{264}{\sqrt{215}}$ cm

Example # 16 : In a $\triangle ABC$, if $8R^2 = a^2 + b^2 + c^2$, show that the triangle is right angled.

Solution : We have : $8R^2 = a^2 + b^2 + c^2$
 $\Rightarrow 8R^2 = [4R^2 \sin^2 A + 4R^2 \sin^2 B + 4R^2 \sin^2 C] \quad [\because a = 2R \sin A \text{ etc.}]$
 $\Rightarrow 2 = \sin^2 A + \sin^2 B + \sin^2 C \Rightarrow (1 - \sin^2 A) - \sin^2 B + (1 - \sin^2 C) = 0$
 $\Rightarrow (\cos^2 A - \sin^2 B) + \cos^2 C = 0 \Rightarrow \cos(A+B) \cos(A-B) + \cos^2 C = 0$
 $\Rightarrow -\cos C \cos(A-B) + \cos^2 C = 0 \Rightarrow -\cos C \{\cos(A-B) - \cos C\} = 0$
 $\Rightarrow -\cos C [\cos(A-B) + \cos(A+B)] = 0 \Rightarrow -2\cos A \cos B \cos C = 0$
 $\Rightarrow \cos A = 0 \text{ or } \cos B = 0 \text{ or } \cos C = 0$
 $\Rightarrow A = \frac{\pi}{2} \text{ or } B = \frac{\pi}{2} \text{ or } C = \frac{\pi}{2}$
 $\Rightarrow \triangle ABC$ is a right angled triangle.

Example # 17 : $\frac{b^2 - c^2}{2a} = R \sin(B - C)$

Solution : $\frac{b^2 - c^2}{2a} = \frac{4R^2(\sin^2 B - \sin^2 C)}{4R \sin A} = \frac{R \sin(B+C) \sin(B-C)}{\sin A} = R \sin(B - C)$



Self Practice Problems :

(12) In a ΔABC , prove that $(a + b) = 4R \cos\left(\frac{A-B}{2}\right) \cos \frac{C}{2}$

(13) In a ΔABC , if $b = 15$ cm and $\cos B = \frac{4}{5}$, find R .

(14) In a triangle ABC if α, β, γ are the distances of the vertices of triangle from the corresponding points of contact with the incircle, then prove that $\frac{\alpha\beta\gamma}{\alpha + \beta + \gamma} = r^2$

Ans. (13) 12.5

Radius of The Incircle :

If 'r' be the inradius of ΔABC , then

(i) $r = \frac{\Delta}{s}$

(ii) $r = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}$

(iii) $r = \frac{a \sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}}$ and so on

(iv) $r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

Radius of The Ex-Circles :

If r_1, r_2, r_3 are the radii of the ex-circles of ΔABC opposite to the vertex A, B, C respectively, then

(i) $r_1 = \frac{\Delta}{s-a}; r_2 = \frac{\Delta}{s-b}; r_3 = \frac{\Delta}{s-c};$

(ii) $r_1 = s \tan \frac{A}{2}; r_2 = s \tan \frac{B}{2}; r_3 = s \tan \frac{C}{2}$

(iii) $r_1 = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}}$ and so on

(iv) $r_1 = 4R \sin \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}$

Example # 18 : $\cos A + \cos B + \cos C = \left(1 + \frac{r}{R}\right)$

Solution :

$$\begin{aligned} \text{LHS} &= \cos A + \cos B + \cos C \\ &= 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) + 1 - 2 \sin^2 \frac{C}{2} \\ &= 2 \sin \frac{C}{2} \left\{ \cos\left(\frac{A-B}{2}\right) - \sin \frac{C}{2} \right\} + 1 = 2 \sin \frac{C}{2} \left\{ \cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right) \right\} + 1 \\ &= 2 \sin \frac{C}{2} \left\{ 2 \sin \frac{A}{2} \sin \frac{B}{2} \right\} + 1 = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &= 1 + \frac{1}{R} \left(4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) = 1 + \frac{r}{R} = \text{RHS} \end{aligned}$$

Example # 19 : In a triangle ABC , find the value of $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3}$.

Solution :

$$\begin{aligned} &\frac{b-c}{\left(\frac{\Delta}{s-a}\right)} + \frac{c-a}{\left(\frac{\Delta}{s-b}\right)} + \frac{a-b}{\left(\frac{\Delta}{s-c}\right)} \\ &= \frac{1}{\Delta} [(b-c)(s-a) + (c-a)(s-b) + (a-b)(s-c)] \end{aligned}$$



$$= \frac{1}{\Delta} [s(b-c+c-a+a-b) - a(b-c) - b(c-a) - c(a-b)] = 0$$

Self Practice Problems :

- (15) In a triangle ABC, r_1, r_2, r_3 are in HP. If its area is 24 cm^2 and its perimeter is 24 cm. then find lengths of its sides.
- (16) In a triangle ABC, $a : b : c = 4 : 5 : 6$. Find the ratio of the radius of the circumcircle to that of the incircle.
- (17) In a ΔABC , prove that $\frac{r_1 - r}{a} + \frac{r_2 - r}{b} = \frac{c}{r_3}$.
- (18) If A, A_1, A_2 and A_3 are the areas of the inscribed and escribed circles respectively of a ΔABC , then prove that $\frac{1}{\sqrt{A}} = \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}}$.

Ans. (15) 6, 8, 10 (16) 16 : 7

Length of Angle Bisectors, Medians & Altitudes :

- (i) Length of an angle bisector from the angle $A = \beta_a = \frac{2bc \cos \frac{A}{2}}{b+c}$;
- (ii) Length of median from the angle $A = m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$
- & (iii) Length of altitude from the angle $A = A_a = \frac{2\Delta}{a}$

NOTE : $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4} (a^2 + b^2 + c^2)$

Example #20 : In ΔABC , AD & BE are its two median. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$ then find the length of BE and area of ΔABC .

Solution : $AP = \frac{2}{3}$; $AD = \frac{8}{3}$; $PD = \frac{4}{3}$; Let $PB = x$

$$\tan 60^\circ = \frac{8/3}{x} \quad \text{or} \quad x = \frac{8}{3\sqrt{3}}$$

$$\text{Area of } \Delta ABP = \frac{1}{2} \times \frac{8}{3} \times \frac{8}{3\sqrt{3}} = \frac{32}{9\sqrt{3}}$$

$$\therefore \text{Area of } \Delta ABC = 3 \times \frac{32}{9\sqrt{3}} = \frac{32}{3\sqrt{3}}$$

$$\text{Also, } BE = \frac{3}{2}x = \frac{4}{\sqrt{3}}$$

Self Practice Problem :

- (19) In a ΔABC if $\angle A = 90^\circ$, $b = 5 \text{ cm}$, $c = 12 \text{ cm}$. If 'G' is the centroid of triangle, then find circumradius of ΔGAB .



Ans. (19) $\frac{13\sqrt{601}}{30}$ cm

The Distances of The Special Points from Vertices and Sides of Triangle :

- (i) Circumcentre (O) : $OA = R$ and $O_a = R \cos A$
- (ii) Incentre (I) : $IA = r \operatorname{cosec} \frac{A}{2}$ and $I_a = r$
- (iii) Excentre (I_1) : $I_1 A = r_1 \operatorname{cosec} \frac{A}{2}$ and $I_{1a} = r_1$
- (iv) Orthocentre (H) : $HA = 2R \cos A$ and $H_a = 2R \cos B \cos C$
- (v) Centroid (G) : $GA = \frac{1}{3} \sqrt{2b^2 + 2c^2 - a^2}$ and $G_a = \frac{2\Delta}{3a}$

Example #21 : If p_1, p_2, p_3 are respectively the lengths of perpendiculars from the vertices of a triangle ABC to the opposite sides, prove that :

(i) $\frac{\cos A}{p_1} + \frac{\cos B}{p_2} + \frac{\cos C}{p_3} = \frac{1}{R}$ (ii) $\frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b} = \frac{a^2 + b^2 + c^2}{2R}$

Solution :

(i) use $\frac{1}{p_1} = \frac{a}{2\Delta}, \frac{1}{p_2} = \frac{b}{2\Delta}, \frac{1}{p_3} = \frac{c}{2\Delta}$

$$\begin{aligned} \therefore \text{LHS} &= \frac{1}{2\Delta} (a \cos A + b \cos B + c \cos C) \\ &= \frac{R}{2\Delta} (\sin 2A + \sin 2B + \sin 2C) = \frac{4R \sin A \sin B \sin C}{2\Delta} \\ &= \frac{4R}{2\Delta} \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} = \frac{1}{4\Delta R^2} abc = \frac{1}{4\Delta R^2} \cdot (4R\Delta) = \frac{1}{R} = \text{RHS} \end{aligned}$$

(ii) $\text{LHS} = \frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b} = \frac{a^2 + b^2 + c^2}{2R} = \frac{2b\Delta}{ac} + \frac{2c\Delta}{ab} + \frac{2a\Delta}{bc} = \frac{2\Delta(a^2 + b^2 + c^2)}{abc}$

$$= \frac{2\Delta(a^2 + b^2 + c^2)}{4\Delta R} = \frac{a^2 + b^2 + c^2}{2R}$$

Self Practice Problems :

(20) If I be the incentre of $\triangle ABC$, then prove that $IA \cdot IB \cdot IC = abc \tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}$.

(21) If x, y, z are respectively be the perpendiculars from the circumcentre to the sides of $\triangle ABC$, then prove that $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{abc}{4xyz}$.

